TED UNIVERSITY
Department of Computer Engineering

CMPE252 MOCK EXAM
SPRING 2025/2026
These are NOT EXAM questions just sample questions

Duration: MM minutes

FULL NAME :

STUDENT ID :

SIGNATURE :

QUESTIONS	POINTS	
Sample Questions 1		/X1
…		/X2
Sample Questions N		/XN
TOTAL		/100

Please Read Carefully

- **This is a closed-book and closed-notes exam.**
- **Talking is strictly forbidden during the exam.**
- **Use of electronic devices including mobile phones is forbidden during the exam.**
- **Students will not be allowed to enter 15 minutes after the exam starts and will not be able to leave in the last 10 minutes of the exam.**
- **Exchange of items (erasers, pens, pencils, etc.) during the exam is strictly forbidden.**
- **There should be a total of X pages. Check your exam paper!**

Name:

Q1 (Multiple Choice)

<u>What could be the output of the c code below?</u>

```
#include <stdio.h>
int main () {
   int  v = 23;
   int  *p;
   p = &v;
   printf("%x\n", &v);
   printf("%x\n", p);
   printf("%d\n", *p);
   return 0;}
```

a.
bffd8b3c
bffd8b3c
23

b.
23
23
bffd8b3c

c.
bffd8b3c
23
bffd8b3c

d.
bffd8b3c
bffd8b3c
bffd8b3c

Q2 (Open Ended)

A polynomial $p(x)$ of degree n is defined as follows:

$$p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots + a_n x^n$$

Write a C program which reads degree and coefficients a_0, a_1, a_2, a_3, …, a_n of a polynomial and evaluates the polynomial at various values of x.

Notes:

- You could use the following function from math library (**#include <math.h>**) which returns **exponent** power of **base**.
  double pow(double base, double exponent);
- Assume that degree of the polynomial could be at most 100.

Name:

Q3. (Fill in the Blanks, Match the following)

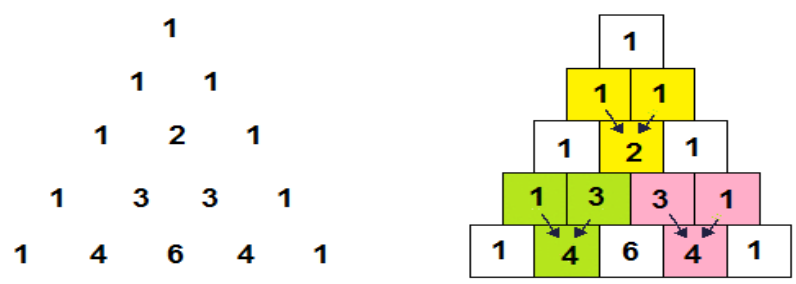
Choose one option from the table and correctly fill in the blanks in the following questions:

indirect reference	<math.h>	error	“stdio.h”	global
pointer direction	prefix	buffer	LIFO	run
pointer	postfix	Place holder	FIFO	Time
linker	infix	overflow	compiler	Free

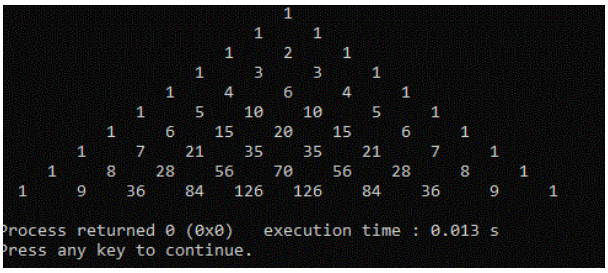
1. The statement `i++;` is called a(n) _____ increment.
2. The preprocessor directive #include _____ should be present in a program that calls the fabs library function.
3. Accessing the contents of memory using a pointer variable that contains the address of that memory location is called _____.
4. If you go beyond the index of a string, you may experience _____ _____.

Q4. Open ended for a specific purpose (USE Recursion)

The following pyramid is called Pascal’s triangle. For n=0, the row has only one element which is 1. Recursively for `n=4`, the elements of the row is `1, (1+3)=4, (3+3)=6, (3+1)=4, 1` as illustrated below. Write <u>the full C code</u> which will CALCULATE Pascal’s triangle for `n=9`, where number of rows is `10`.



Sample execution:



HINT: Each cell is a Binomial coefficient and should satisfy a recursion rule